CIARA: Integrating AI in a Low-Cost Robotic Arm for Image-to-Sketch Reproduction

MINI PROJECT REPORT

Submitted by

Albin Thomas (KTE22RAI006)
Alex Chacko (LKTE22RAI034)
Tony Basil Babu (KTE22RAI033)
Muhammed Shahsad A.P (KTE22RAI020)

to

APJ Abdul Kalam Technological University
in partial fulfilment of the requirements for the award of Degree of

BACHELOR OF TECHNOLOGY

in

ROBOTICS AND ARTIFICIAL INTELLIGENCE

Department Of Electronics and Communication Engineering
RAJIV GANDHI INSTITUTE OF TECHNOLOGY

KOTTAYAM - 686501
April 2025

Department Of Electronics and Communication Engineering
RAJIV GANDHI INSTITUTE OF TECHNOLOGY
KOTTAYAM - 686501

BACHELOR OF TECHNOLOGY

Certificate

This is to certify that the Mini project entitled **Project Title** is a bonafide work carried out by Albin Thomas(KTE22RAI006), Alex Chacko(LKTE22RAI034), Tony Basil Babu(KTE22RAI033), and Muhammed Shahsad A.P(KTE22RAI020) during 2025-26, in partial fulfilment for the award of the B. Tech degree in Robotics and Artificial Intelligence of APJ Abdul Kalam Technological University, Rajiv Gandhi Institute of Technology, Kerala.

Guide

Dr. ANILKUMAR C.D.

Prof. NELSA ABRAHAM

Co-ordinators

Prof. [Coordinator Name 1]

Prof. [Coordinator Name 2]

Head of the Department, ECE

Dr. DAVID SOLOMON GEORGE

ACKNOWLEDGEMENT

We extend our heartfelt gratitude to **DR. Anil Kumar C D** and **Prof. Nelsa Abraham** for their invaluable guidance, encouragement, and unwavering support throughout this project. Their insights and expertise have been instrumental in shaping our work and bringing our ideas to fruition.

This project was made possible through the collective effort of our entire team. Each member has played a crucial role in various aspects, from conceptualization and design to implementation and testing. Their dedication and teamwork have been vital to the successful completion of this endeavor.

We are profoundly grateful to **Dr. David Solomon George**, Head of the Department of Electronics and Communication Engineering, for granting us permission and ensuring that we had access to the necessary resources and facilities to carry out this project effectively. We also express our sincere appreciation to **Dr. Prince A**, our Principal, for providing the infrastructure and support required to see this project through.

Additionally, we wish to acknowledge the contributions of the teaching and non-teaching staff of the Department of Electronics and Communication Engineering for their continuous assistance and cooperation.

Lastly, we extend our deepest thanks to our friends, family, and well-wishers for their encouragement, guidance, and unwavering support, which have motivated us to complete this project successfully.

ABSTRACT

Robotic drawing systems have garnered significant interest for their ability to replicate human-like artistic skills. In this paper, we present a low-cost robotic arm drawing system that combines machine learning and custom algorithms to generate and reproduce sketches with high accuracy.
Our system utilizes **Pollinations.ai** to fetch and preprocess input images, which are then processed using **CycleGAN** to produce a sketch. This sketch is further refined using a **skeletonization** algorithm and a **custom sketching algorithm** to generate precise waypoints. A 5-degree-of-freedom (5-DOF) robotic arm equipped with servo motors follows these waypoints to recreate the sketch on a physical canvas. Inverse kinematics, solved and implemented using **MATLAB**, ensures accurate motion control of the robotic arm.
One of the key challenges we faced was the limited precision and control of the servo motors due to the arm's low cost (under $100), which affected the quality of the output. Despite these hardware limitations, the system demonstrated impressive results in simulation, highlighting the effectiveness of our approach. The robotic arm was successfully showcased at our college's mini project expo, where it demonstrated the ability to produce complex sketches with reasonable accuracy.
This work highlights the potential of low-cost robotic systems in creative applications, offering a foundation for future improvements in precision and performance.

KEYWORDS: Robotic Drawing Arm, **CycleGAN**, **Custom Sketching Algorithm**, **AI**, **Robotics**

LIST OF CONTENTS

1. INTRODUCTION

Robotic drawing systems serve as a bridge between technology and creativity, allowing machines to replicate human-like artistic skills. These systems have applications in art, design, and education, but achieving high precision while maintaining affordability remains a significant challenge. The reliance on expensive hardware and complex algorithms often limits accessibility.

This paper presents a cost-effective robotic arm drawing system capable of generating and reproducing sketches with notable accuracy. The system integrates machine learning and custom algorithms, utilizing *Pollinations.ai* for image preprocessing and *CycleGAN* for sketch transformation. A *skeletonization* technique and a *custom sketching algorithm* refine the sketches into precise waypoints. A 5-degree-of-freedom (5-DOF) robotic arm with servo motors then follows these waypoints to recreate the sketches. To enhance motion accuracy, inverse kinematics is implemented using MATLAB.

While the use of budget-friendly servo motors (costing under $120) introduces some precision limitations, the system demonstrated strong performance in simulations. It was successfully showcased at a college project expo, where it produced detailed sketches with reasonable accuracy. This work highlights the potential of combining machine learning with low-cost robotics, laying the foundation for future advancements in accessible robotic drawing technology.

2. OBJECTIVES

- **Develop a low-cost robotic drawing system:** Design and implement a 5-degree-of-freedom (5-DOF) robotic arm using budget-friendly components
- **Integrate machine learning for sketch generation:** Utilize *Pollinations.ai* for image preprocessing and employ *CycleGAN* to convert input images into sketches for the robotic arm to replicate.
- **Implement path planning and control algorithms:** Develop a skeletonization technique and a custom sketching algorithm to extract precise waypoints for the robotic arm's movements.
- **Enhance motion accuracy using inverse kinematics:** Apply inverse kinematics solutions in *MATLAB* to improve the precision and efficiency of the robotic arm's drawing capabilities.
- **Validate performance through simulation and real-world testing:** Assess the system's effectiveness by evaluating its performance in both simulated environments and physical demonstrations.

3. DESIGN PARAMETERS

3.1 Workspace and Task Planned

3.1 Workspace Considerations

- The arm covers a **semi-circular workspace** with a **34 cm reach radius** from its base.
- The arm operates on a **flat plane**, enabling precise **portrait drawing**.
- The **end effector is a pen**, designed for smooth and accurate strokes.
- A **3D-printed stabilizer** minimizes jerky movements caused by motor vibrations.
- The design ensures **clearance from obstructions** to prevent mechanical interference.

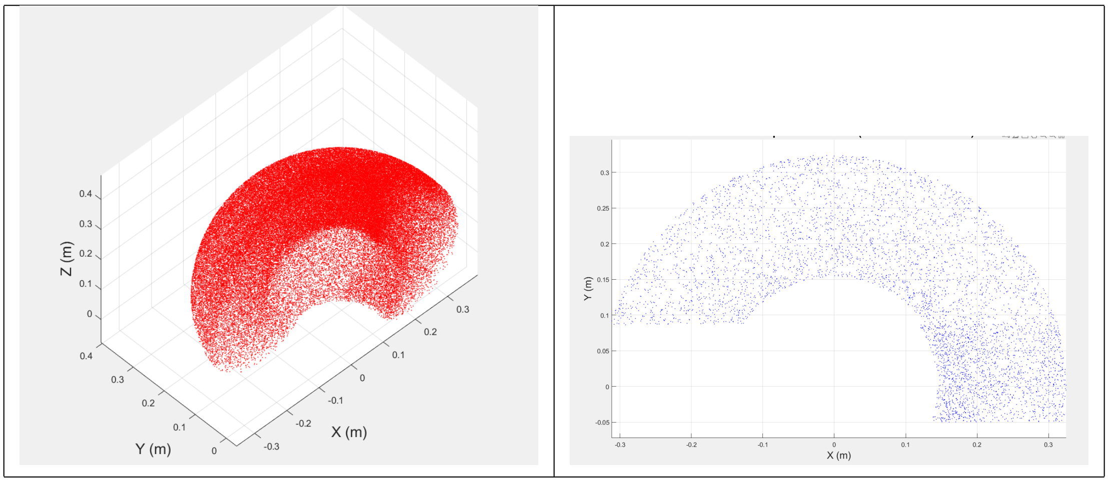


Table 1: Workspace Simulation From Matlab

3.1.2 Task Planned

- Precise portrait drawing within the **limited workspace**.
- **180-degree rotational base (DS3240 servo)** allows drawing in the **first and second quadrants**.
- **Smooth horizontal and vertical motion** ensures accurate stroke placement.
- **Pen stabilization mechanism** minimizes vibrations for clean, continuous lines.
- **Optimized motion planning** prevents overlapping and ensures proportional sketches.
- **Efficient path execution** reduces unnecessary movements for faster drawing.
- **Obstacle-free design** eliminates mechanical interference during operation.

3.2 Structural Parameters and Drawing

3.2.1 Material Used

- The materials used in constructing the robotic arm are selected based on affordability, strength, and ease of fabrication:
- **PVC Pipes** – Used for the main structural frame to provide a lightweight and cost-effective solution.
- **3D Printed Parts** – Custom-designed joints and connectors for precise movement.
- **Metal Screws and Wires** – Used for assembling components securely.
- **Servo Motors** – Various servo motors with different torque capacities are utilized for joint actuation.

3.2.2Number of Joints and Type of Joints

Joint No.	Type of Joint	Actuator Used	Material
Base Joint (J1)	Revolute (Rotational)	Servo Motor (20 kg-cm)	PVC & 3D Printed Parts
Shoulder Joint (J2)	Revolute (Rotational)	Servo Motor (40 kg-cm) DS3240	PVC & 3D Printed Parts
Elbow Joint (J3)	Revolute (Rotational)	Servo Motor (40 kg-cm) DS3240	PVC & 3D Printed Parts
Wrist Joint (J4)	Revolute (Rotational)	Servo Motor (9 kg-cm) MG996R	PVC & 3D Printed Parts

Table 2: Joints and Actuation Details of the Robotic Arm

3.2.3 Link Length and Structure

Link No.	**Length (cm)**	**Material**
L1 (Base to link 1)	4 cm	PVC Pipe
L2 (link 1 to link 2)	16.51 cm	PVC Pipe
L3 (Link 2 to Link 3)	16.4 cm	PVC Pipe
L4 (Link 3 to end effector)	11.5 cm cm	PVC Pipe

Table 3: Link Lengths and Materials

3.2.4 Actuation and Control

- The robotic arm is actuated using 5 **servo motors** of different torque ratings.
- A **16-channel servo motor driver** is used to control the movement of the robotic arm.
- A **Raspberry Pi 4** acts as the main controller to process commands and execute tasks.

3.2.5 Robotic Arm CAD Model

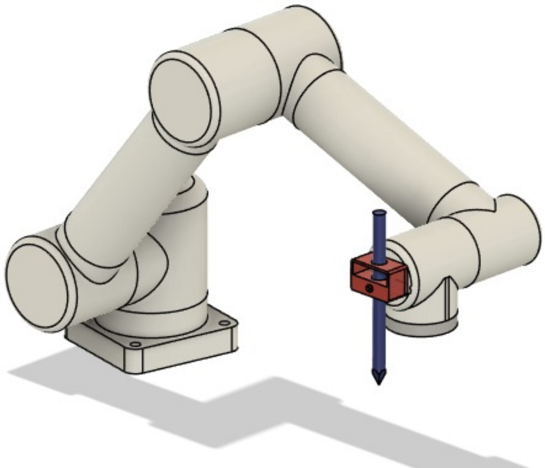


Figure 1: Schematic Diagram of the Robotic Arm

3.2.5.1 Pen Holder

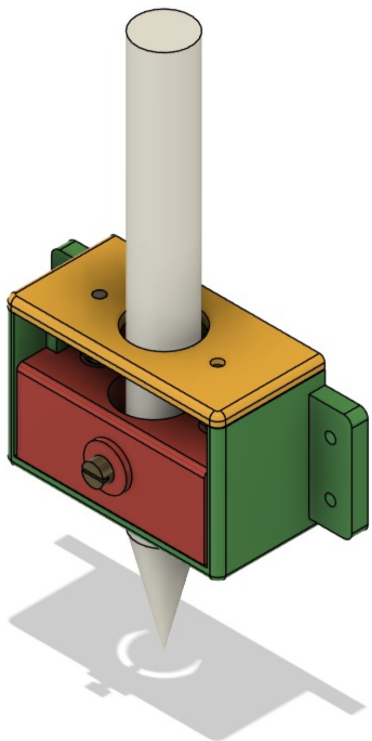


Figure 2: Schematic Diagram of the Pen Holder

3.3 Mechanics/ Mechanical Parameters

Joint (i)	θ_i **(rad)**	d_i **(cm)**	a_i **(cm)**	α_i **(rad)**
1	0	4.0	0	0
2	1.5708	8.3609	0	1.5708
3	3.1416	7.4	0	0
4	3.1416	1.3	0	0

Table 4: Denavit-Hartenberg Parameters (Standard Convention)

Joint Motion Analysis: Position, Velocity, Acceleration, and Torque

- This section presents the dynamic analysis of the robotic arm as it moves from its **home position** $[0°, 0°, 0°, 0°]$ to the **target configuration** $[0°, 90°, 0°, 0°]$. The graphs illustrate:
- **Joint Position**: The change in angular displacement over time.
- **Joint Velocity**: The rate of change of position, showing smooth transitions.
- **Joint Acceleration**: The variation of velocity, highlighting stability and motion efficiency.
- **Joint Torque**: The required force at each joint to achieve the desired movement.

These plots provide insight into the **kinematic and dynamic behavior** of the arm, ensuring **controlled and stable operation** during execution.

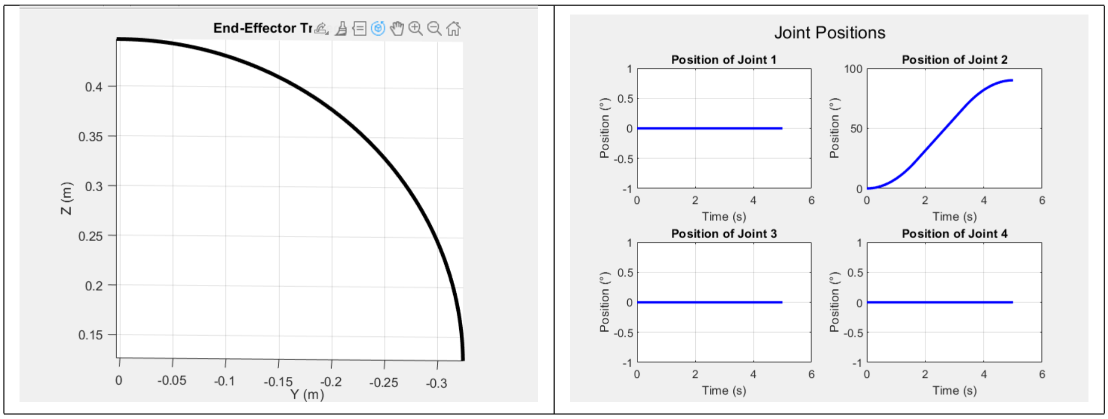


Table 5: End Effector,Joint Position Vs Time

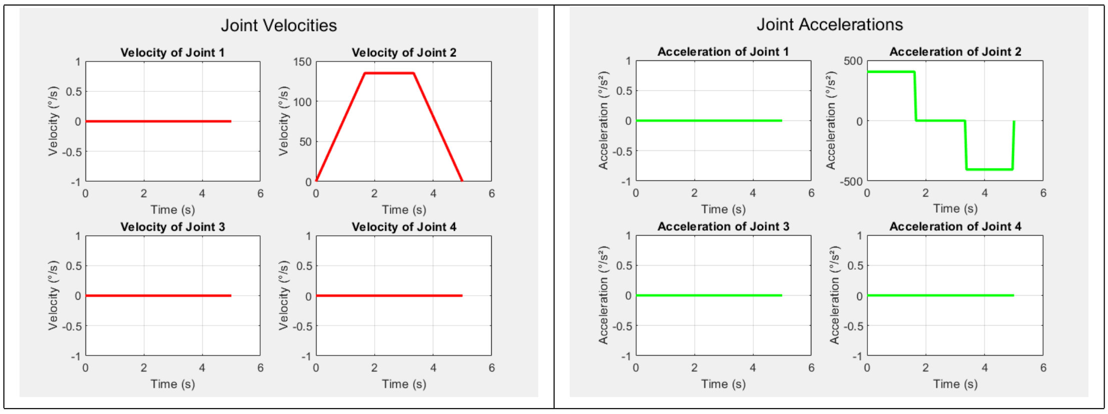


Table 6: Velocity,Acceleration Vs Time

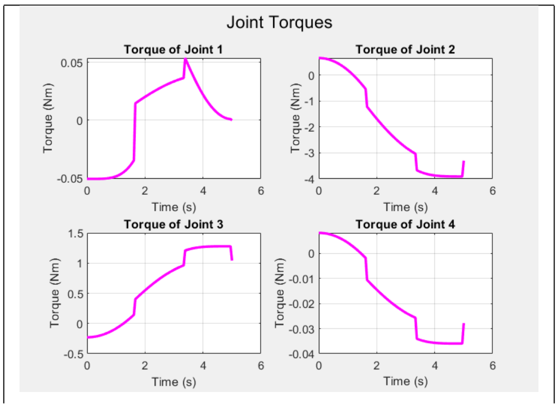


Table 7: Torque Vs Time

3.4 Electrical/ Electronics Control

3.4.1 WorkFlow

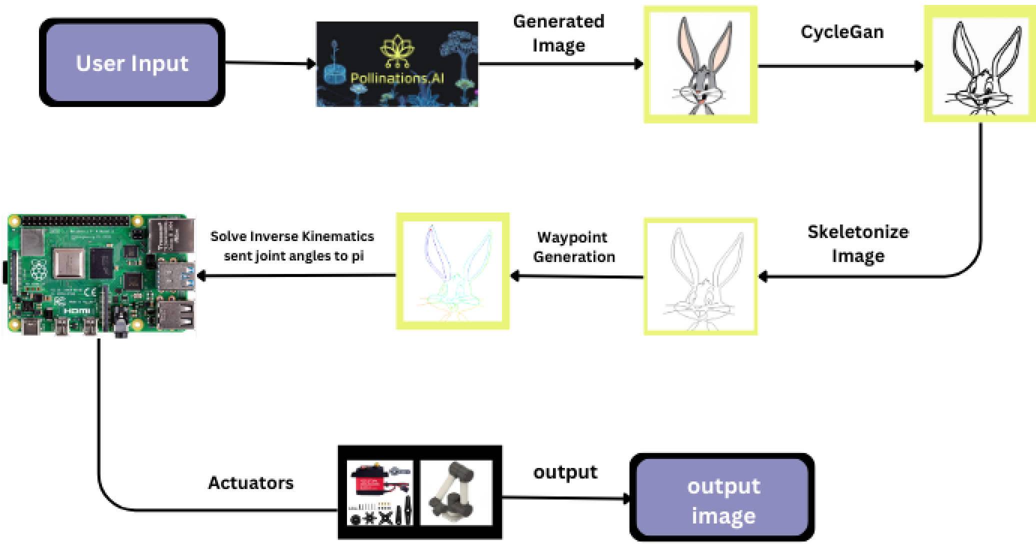


Figure 3: Work Flow Block Diagram

1. User Input and Image Generation

- The process begins with **user input**, which can be a text prompt or an initial image.
- This input is processed by **Pollinations.AI**, an AI-based image generation tool that creates an image based on the given prompt.

2. Image Processing using CycleGAN

- The **generated image** is passed through **CycleGAN**, a deep learning model for *image-to-image translation*.
- This step refines the generated image into a stylized version, making it suitable for further processing.

3. Skeletonization and Waypoint Generation

- The stylized image is converted into a **skeletonized version**, which extracts the primary features.
- Using the skeletonized image, **waypoints** are generated to define the trajectory that the robotic system will follow.

4. Raspberry Pi Control and Inverse Kinematics

- The waypoints are processed to compute the required **joint angles** for the robotic arm.
- These joint angles are sent to a **Raspberry Pi**, which acts as the primary controller.
- The Raspberry Pi **solves inverse kinematics** to ensure precise movement along the planned path.

5. Actuator Control and Output Generation

- The Raspberry Pi **controls actuators** such as servo motors to replicate the image physically.
- The robotic system follows the planned motion to create the **final output image**.

3.4.2 Circuit Diagram

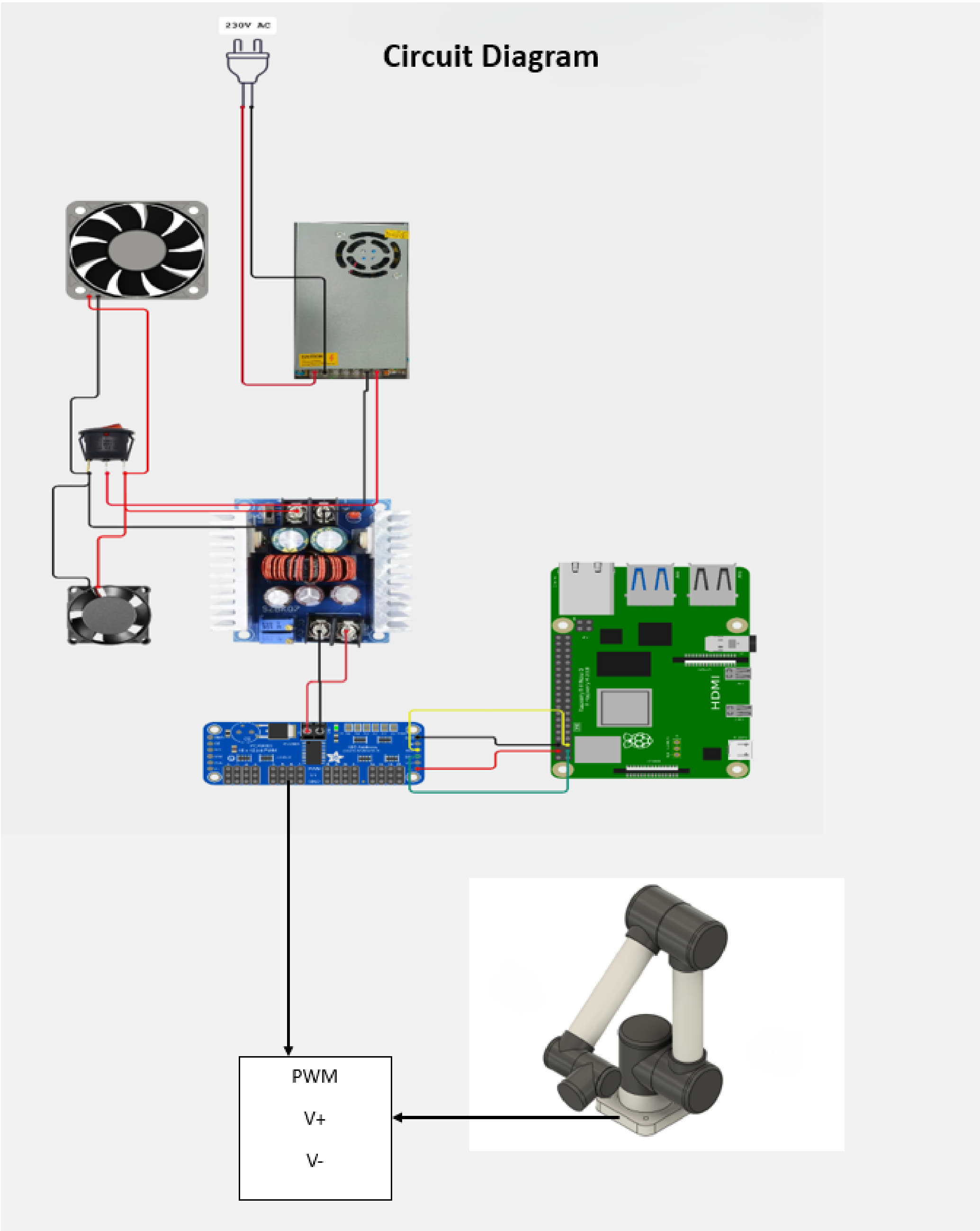


Figure 4: Work Flow Block Diagram

3.4.3 Electrical Components

Raspberry Pi 4

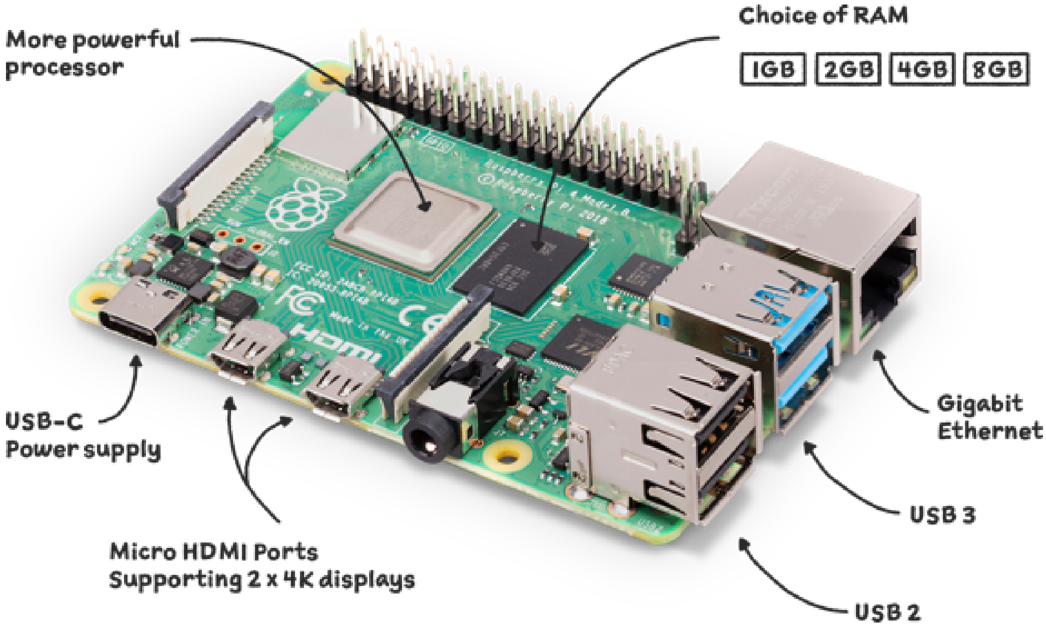


Figure 5: Work Flow Block Diagram

The **Raspberry Pi 4** is a high-performance SBC for IoT, robotics, and AI. It features a **quad-core Cortex-A72 CPU**, up to **8GB RAM**, and multiple connectivity options.

- **Fast CPU & RAM**
- **USB 3.0 & Dual HDMI**
- **WiFi, Bluetooth, microSD, SSD**
- **Supports Linux, Python, ROS, AI**

Ideal for **IoT, automation, robotics, and AI** with low power use and strong community support.

Servo Motors

DS3240 Servo Motor (40 kg-cm)

Figure 6: Work Flow Block Diagram

The **DS3240** is a high-torque servo motor capable of providing up to 40 kg-cm of torque. It was chosen for applications that require handling significant loads while maintaining stability. This motor ensures strong grip force and smooth movement, making it ideal for robotic arms and heavy-duty mechanisms.

20 kg-cm Servo Motor

Figure 7: Work Flow Block Diagram

This medium-torque servo motor was used in sections of the project where moderate force was required. It provides a balance between power consumption and torque, making it a reliable option for controlled movements.

MG996R Servo Motor (9 kg-cm)

Figure 8: Work Flow Block Diagram

The MG996R is a budget-friendly servo motor with a 9 kg-cm torque rating. It was used in areas that require lightweight movement and low power consumption. Although it is not as strong as the other motors, it is a cost-effective choice for tasks that do not require high torque.

16-Channel Servo Motor Driver

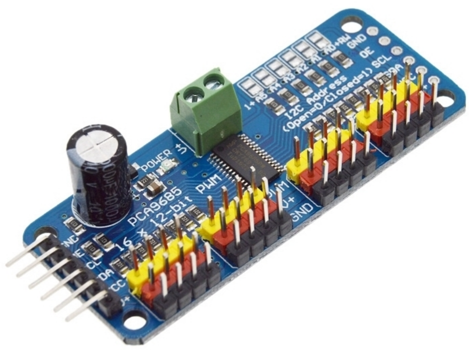


Figure 9: Work Flow Block Diagram

To efficiently control multiple servo motors, we used a 16-channel servo motor driver. This driver allows for precise control over all the motors while ensuring smooth and synchronized motion.

Details of Cost Involved

Items	**Quantity**	**Cost in INR**
PVC Pipe	2.5 inch, 3.5 inch	520/-
Servo motor (40 kg-cm) DS3240	2	2400/-
Servo motor (20 kg-cm)	1	1000/-
Servo motor (9 kg-cm) MG996R	1	300/-
Servo motor Driver 16 channel	1	150/-
3D printed parts	–	200/-
Wire (Single and Multi Strand)	1m	85/-
Screw	2mm × 8mm	250/-
Raspberry Pi 4	1	4500/-
Pen	3	30/-
Power Supply	1	600/-
Buck Converter	1	200/-
		Total = 10235/-

Table 8: Total Cost

Results

Robotic Arm

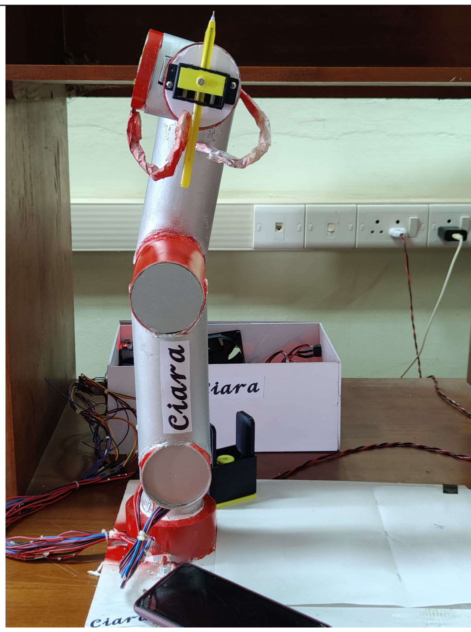
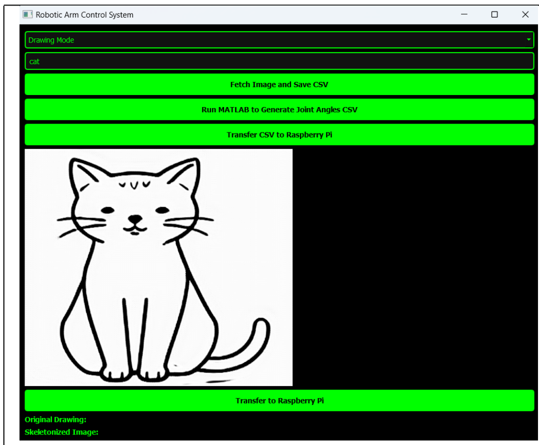


Table 9: Robotic Arm

Simulation Results



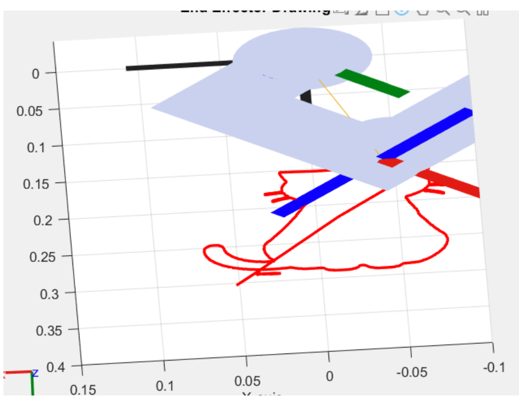


Table 10: Simulation Result With Ideal Conditions(Time Taken: 3 Mins)

Drawing Results Vs Time Taken

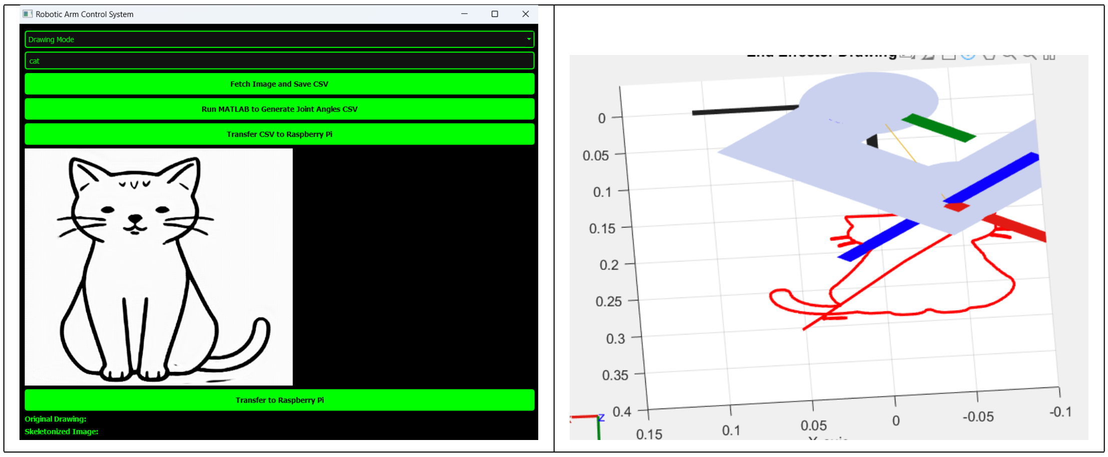


Table 11: Simulation Result With Ideal Conditions(Time Taken: 3 Mins)